MTH 310 **Quiz 5** Name:______________________

Spring, 2026

Instructions. (15 points) Sections 14.6–14.7. Show relevant work for full credit.

(5pts) **1.** Let $f(x, y) = x^2 y + \sqrt{y}$.

(a) *(3 pts)* Find the directional derivative of f at the point $(2, 1)$ in the direction toward the point $(5, 5)$.

(b) *(2 pts)* What is the minimum rate of change of f at $(2, 1)$, and in what direction does it occur?

(4pts) **2.** Find all critical points of $f(x, y) = x^3 - 6xy + 8y^3$ and classify each as a local maximum, local minimum, or saddle point.

9 pts

(6^{pts}) **3.** Find the absolute maximum and absolute minimum values of $f(x, y) = x^2 - 2x + y^2$ on the closed disk $x^2 + y^2 \leq 4$.

6 pts